

David Kreuter

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Senior Infrastructure & Network Automation Engineer

Infrastructure automation architect with 10+ years of experience building production-grade orchestration platforms, API-driven provisioning systems, and network automation at enterprise scale. Expert in Python-based middleware development, infrastructure-as-code (Terraform, Ansible), and API design for lifecycle management of 20,000+ virtual machines across geo-redundant data centers. Extensive experience in network security automation (firewall rule generation, DMZ provisioning, IP management), high-availability architecture, and building compliant infrastructure for a regulated financial institution. Proven ability to design scalable provisioning engines, integrate complex multi-vendor APIs, and mentor engineering teams on GitOps best practices.

WORK EXPERIENCE

M&T Bank

03/2019 - Present

Automation Engineer (formerly Systems Engineer)

- **Network Provisioning & Lifecycle Management:** Developed an intelligent IP provisioning engine via InfoBlox API integrations with automated subnet allocation and network resource lifecycle management, eliminating 1,000+ annual engineer hours and resolving a multi-year infrastructure bottleneck.
- **Network Security Automation:** Automated Check Point firewall rule generation and DMZ provisioning; integrated security policy updates into the server build cycle to ensure day-zero network isolation and a hardened security posture across production environments.
- **Platform Engineering & Middleware:** Designed and developed a custom Python orchestration application as middleware between ServiceNow and Ansible Automation Platform; implemented advanced retry logic, auto-remediation of deployment failures, and a pub/sub queue model for high-volume scaling.
- **Architectural Scale:** Engineered a global server provisioning framework that deployed and managed 20,000+ virtual machines; integrated Rubrik and VMware APIs to automate backup and compute placement, ensuring 100% RTO/RPO compliance at build time.
- **Network APIs & Integration:** Developed robust REST API integrations across VMware, Rubrik, InfoBlox, ServiceNow, and Check Point platforms, enabling programmatic network and infrastructure provisioning with full lifecycle management from deployment through decommissioning.
- **Infrastructure Resilience:** Engineered and maintained a highly resilient infrastructure architecture spanning three identical, geo-redundant data centers with automated failover to ensure maximum uptime and disaster-recovery readiness.
- **Hybrid Cloud & GitOps:** Architected GitOps-driven CI/CD pipelines using Terraform to automate the deployment of virtual machines to Azure; implemented a pull-request-based workflow that standardized infrastructure changes through automated plan/apply cycles.
- **Cloud Abstraction & Redesign:** Refactored legacy on-premises automation workflows to be fully cloud-agnostic, enabling seamless deployment to Azure; designed the logic to be transparent to end users, maintaining a consistent interface while abstracting the underlying infrastructure provider.
- **End-to-End Hardening & Provisioning:** Developed a zero-touch build workflow that delivers fully patched and hardened Linux (Ansible) and Windows (GPO) servers; automated agent installation, drive formatting, and optional hardened SQL deployments at scale.
- **Database as a Service (DBaaS):** Collaborated with Database Engineering to architect a high-performance on-premises SQL cluster, enabling a unified DBaaS offering that consolidated fragmented legacy databases and streamlined application delivery.
- **Technical Leadership:** Serve as a primary mentor for the Technology Development Program (TDP), setting enterprise-wide standards for naming conventions, Git branching strategies, and unit testing for infrastructure code.
- **Cost Optimization:** Orchestrated the migration of 850 branch file servers within a strict three-month window using custom PowerShell automation, avoiding \$400,000 in extended licensing fees.
- **Risk Mitigation:** Authored Active Directory lifecycle scripts with built-in "Read-Only" dry-run modes; implemented least-privilege permissions and robust rollback protocols for stale object cleanup.

OTG Management

07/2013 - 02/2019

Lead Network Engineer

- **Infrastructure Resilience:** Architected a highly redundant DNS solution that significantly reduced network outages and optimized global traffic performance across a distributed multi-site environment.
- **Cloud Connectivity:** Established secure, encrypted VPN tunnels to bridge on-premises networks with cloud resources, ensuring seamless hybrid-cloud operations.
- **Performance Engineering:** Diagnosed and resolved a mission-critical server hang by identifying a NUMA architecture mismatch; realigned vCPU/Memory topology to eliminate cross-node latency and restore system stability.
- **Security Architecture:** Engineered a secure reverse proxy solution to protect sensitive internal infrastructure, enabling customer online ordering while mitigating external threat vectors.

Network Engineer

- **Network Administration:** Managed complex firewall rules and switch ACLs for a distributed network spanning 12 airports and 3 satellite offices, enforcing tenant-level isolation and security policy compliance.
- **Proactive Monitoring:** Developed service-aware alerting and telemetry frameworks for business-critical systems, reducing mean time to resolution (MTTR) through proactive incident detection and enhanced network observability.
- **Cybersecurity Governance:** Directed enterprise-wide vulnerability scanning and remediation efforts for internal and external systems to maintain a hardened security posture.
- **Enterprise Mobility:** Onboarded and managed 15,000 iPads via Meraki MDM, implementing automated security protocols including geo-fencing, remote-wipe triggers, and custom application deployment.

Systems Administrator

- **Disaster Recovery:** Spearheaded the deployment of Datto backup and disaster-recovery appliances to ensure 100% business continuity for critical operations.
- **Operational Visibility:** Modernized infrastructure monitoring by designing custom Grafana dashboards to visualize real-time performance metrics and network utilization.

SKILLS

Languages: Python, Bash, SQL, PowerShell, YAML, Javascript, PHP

Networking: BGP, DNS, VPN, F5 Load Balancing, High Availability, VRRP, Intrusion Detection, Vulnerability Scanning, Meraki MDM

Automation/Orchestration: Terraform, REST API, GraphQL, GitOps, CI/CD Pipelines, Ansible Automation Platform, Middleware Development, Gitlab, Grafana, ServiceNow, Unit Testing

Infrastructure: Infrastructure as Code (IaC), Check Point Firewalls, VMware vSphere, Azure Cloud Services, OS Hardening (CIS/STIG), NUMA Tuning, Nutanix AHV, Rubrik, Active Directory, DBaaS

EDUCATION

Bachelor Of Science in Computer Information Systems

12/2012

SUNY Fredonia

CERTIFICATIONS

Certified Meraki Network Operator

Meraki

VOLUNTEERING & LEADERSHIP

Habitat For Humanity

Volunteer

07/2025 - Present

Tift Nature Preserve

Volunteer

01/2025 - Present

716 Paws

Thermal drone pilot for search and rescue missions.

01/2022 - Present